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ACTIVITY A AND SUBMISSION.

ACTIVITY A

Integer.parseInt(“ABC”);

String name = null; name.length();

New java.io.FileReader(“missing.txt”);

Integer.parseInt("ABC");

Exception, NumberFormatException.

Classification, Unchecked exception.

Explanation: This exception occurs because `"ABC"` contains letters that cannot be converted into an integer.

Handling strategy: Use a try-catch block to catch NumberFormatException.

```
String name = null; name.length();
```

- Exception: NullPointerException.
- Classification: Unchecked exception.
- Explanation: This exception occurs because the variable `name` has a null value, meaning it does not reference a String object. Therefore, Java cannot call the `length()` method on it.
- Handling strategy: Check whether the variable is null before calling its method

```
new java.io.FileReader("missing.txt");
```

- Exception: FileNotFoundException.
- Classification: Checked exception.
- Explanation: This exception occurs when Java tries to open a file that does not exist or cannot be accessed.
- Handling strategy: Check that the file exists and use a try-catch block to handle FileNotFoundException.

4. OutOfMemoryError occurs when the Java Virtual Machine runs out of memory needed to execute the program. Catching this error and continuing may cause further failures because the application may be in an unstable state.

ACTIVITY D

Q1. FileNotFoundException and IOException are checked exceptions. The `throws` keyword appears in the method declaration.

Q2. `throw` is used to explicitly create and raise an exception inside a method.

throw new IllegalArgumentException("Mark must be between 0 and 100");

`throws` is used in a method declaration to indicate that the method may pass an exception to the calling method.

static void importMarks(String filename) throws IOException

Q3. A friendly message helps users understand what went wrong without confusing them with technical programming details. The logger records detailed information about the exception to help developers identify, investigate and fix problems.

Q4. An empty catch block silently ignores exceptions, making it difficult to identify and fix errors. Catching Exception before FileNotFoundException makes the specific catch block unreachable because Exception is a superclass of FileNotFoundException. This results in a compilation error.

CONSOLE OUTPUTS FOR T1-T5

PS C:\Users\HP\Documents\ExceptionLab> javac MarksImporter.java DuplicateStudentException.java

PS C:\Users\HP\Documents\ExceptionLab> java MarksImporter

Accepted: ICT001 - 78

Reading: ICT001, 78

Invalid numeric mark for ICT002: ABC

Nov 09, 2026 8:50:11 PM MarksImporter importMarks

WARNING: NumberFormatException: For input string: "ABC"

Duplicate record skipped: Student ID already exists: ICT001

Nov 09, 2026 8:50:11 PM MarksImporter importMarks

WARNING: DuplicateStudentException: Student ID already exists: ICT001

Accepted: ICT003 - 90

Reading: ICT003, 90

Import attempt finished

PS C:\Users\HP\Documents\ExceptionLab> java MarksImporter

Accepted: ICT001 - 78

Reading: ICT001, 78

Invalid numeric mark for ICT002: ABC

Nov 09, 2026 8:54:35 PM MarksImporter importMarks

WARNING: NumberFormatException: For input string: "ABC"

Duplicate record skipped: Student ID already exists: ICT001

Nov 09, 2026 8:54:35 PM MarksImporter importMarks

WARNING: DuplicateStudentException: Student ID already exists: ICT001

Accepted: ICT003 - 90

Reading: ICT003, 90

Invalid mark for ICT004: Mark must be between 0 and 100

Nov 09, 2026 8:54:35 PM MarksImporter importMarks

WARNING: IllegalArgumentException: Mark must be between 0 and 100

Import attempt finished

PS C:\Users\HP\Documents\ExceptionLab> javac MarksImporter.java DuplicateStudentException.java

>> java MarksImporter

Marks file not found; check the file name

Nov 09, 2026 8:59:18 PM MarksImporter main

SEVERE: FileNotFoundException: absent.txt (The system cannot find the file specified)

Import attempt finished

PS C:\Users\HP\Documents\ExceptionLab> javac MarksImporter.java DuplicateStudentException.java

PS C:\Users\HP\Documents\ExceptionLab> java MarksImporter

Accepted: ICT001 - 78

Reading: ICT001, 78

Invalid numeric mark for ICT002: ABC

Nov 09, 2026 9:01:55 PM MarksImporter importMarks

WARNING: NumberFormatException: For input string: "ABC"

Duplicate record skipped: Student ID already exists: ICT001

Nov 09, 2026 9:01:55 PM MarksImporter importMarks

WARNING: DuplicateStudentException: Student ID already exists: ICT001

Accepted: ICT003 - 90

Reading: ICT003, 90

Invalid mark for ICT004: Mark must be between 0 and 100

Nov 09, 2026 9:01:55 PM MarksImporter importMarks

WARNING: IllegalArgumentException: Mark must be between 0 and 100

Import attempt finished

SKIP AND CONTINUE.

I chose to use the skip-and-continue approach in my marks importer. Meaning when the program encounters an invalid record, such as a non-numeric mark, an out-of-range mark or a duplicate student ID, it reports the problem and skips that record instead of stopping the entire import process. The program then continues reading the remaining records in the file. I chose this approach because one incorrect record should not prevent other valid student records from being

processed. This makes the program more reliable and allows valid marks to be imported even when some records contain errors.